SYNTAX ERROR 2026
OCT23-OCT25

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RadixArk
and
Lexsi.ai

PROBLEM STATEMENT - LOCAL LLM TRACK

Track Core Requirement:
All submissions under the Local LLM Track must use SGLang as the core inference and serving engine for model deployment, evaluation, or benchmarking.

Ecosystem Toolkits:
For data curation, fine-tuning, and safety evaluations, participants are to use Lexsi Labs' open-source alignment toolkits across the post-training pipeline.

Participant Guideline:
Teams competing in this track must select **ONE** problem statement from the options below to attempt.

Problem Statements:

1. **Domain assistants for regulated industries:** Adapt a model to a regulated domain such as banking, insurance, legal or healthcare, where answers must follow policy and stay accurate.
2. **Replacing a closed-model API:** Take a high-volume business task that is usually sent to a large closed model (classification, extraction, summarization, drafting) and fine-tune a small open model to match it at a fraction of the cost.
3. **Grounded answers and fewer hallucinations:** Build a training set from an organisation's own documents (manuals, policies, contracts, research) and train a model that answers only from them. It should say so when the answer is not there and make measurably fewer unsupported or false claims.

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4. **Low-resource and multilingual languages:** Improve a model for underserved languages, dialects or code-mixed text in a public-facing setting such as education, public services, agriculture or healthcare.
5. **Structured output and extraction:** Turn unstructured text (forms, invoices, reports, clinical notes, support tickets) into reliable structured data that passes schema validation.
6. **Reasoning with verifiable answers:** Improve step-by-step reasoning in areas where answers can be checked automatically, such as math, financial calculations, code or logic, using reward-based training.
7. **Preference and style alignment:** Align a model with an organization's quality bar or house style (customer communication, technical writing, feedback on student work) using preference data.
8. **Data-scarce domains:** Where little real data exists, generate, verify and filter synthetic training data, and show that curation measurably beats training on raw generated data.
9. **Long documents:** Fine-tune for summarising and reasoning over long documents such as filings, contracts, transcripts or research papers through multi-stage fine tuning.

Stages:
Stage 1 : Data curation (CuratorKIT uses SGLang internally)
Stage 2 : Finetuning (Aligntune uses unsloth/TRL)
Stage 3 : Evals / Inference (AuditKIT can use SGLang for evals), SGLang for inference

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Developer Resources

Tool/Library	Documentation Link	Source
SGLang	https://docs.sglang.io/	https://github.com/sgl-project/sglang
CuratorKit	https://curatorkit.lexsi.ai/latest/	https://github.com/Lexsi-Labs/CuratorKIT
AlignTune	https://aligntune.lexsi.ai	https://github.com/Lexsi-Labs/aligntune
AuditKit	https://auditkit.lexsi.ai	https://github.com/Lexsi-Labs/AuditKIT